

Predicate Logic Proofs

30 March 2026

Housekeeping

- No labs this week, due to the university “reading day” on Friday.
- Extra credit: Due before 5:00PM Thursday, in your TA’s mailbox on the third floor of the YMCA building. (When you enter the mail/copy room, the mailboxes are on the left.
- For next time, try completing the following exercises from 3.3:
 - S88, S89, S90, and if feeling ambitious S91.

Why proofs?

- As a reminder, proofs are often called “systems of *natural deduction*” because they emulate the way we reason through arguments naturally. Studying proofs, i.e. how arguments “work,” pays off dividends in allowing us to “see” how we navigate from true premises to a true conclusion, which in turn demonstrates “how” the argument is valid.
- Truth preservation is what the rules do. They ensure that when fed true inputs, they provide true outputs.



Vocabulary side
quest!

Open Formula

An **OPEN FORMULA** is the result of replacing at least one occurrence of a name in a WFF by a new variable (one not already occurring in the WFF).

(Open formulas are not WFFs, and hence never appear as sentences in proofs. The notion of an open formula is used to present the rules of proof for predicate logic.)

Going from Fa to Fx

Scope

The **SCOPE** of a quantifier in a given formula is the shortest open formula to the right of the quantifier. For example, in the following:

$$(\forall x Fx \ \& \ \exists y (Gy \ \& \ Hy))$$

The scope of the $\forall x$ is Fx

and the scope of the $\exists y$ is $(Gy \ \& \ Hy)$

A quantifier whose scope contains another quantifier is said to have **WIDER SCOPE** than the second. The second is said to have **NARROWER SCOPE** than the first.

$$\forall x(Fx \rightarrow \exists yGy)$$

In this example the $\forall x$ has wider scope than the $\exists y$.

Universalization

A **UNIVERSALIZATION** of a sentence with respect to a given name occurring in the sentence is obtained by the following two steps:

1. Replace all occurrences of the name in the sentence by a variable α , where α does not already occur in the sentence.
2. Prefix $\forall\alpha$ to the open formula resulting from step 1.

Existentialization

An **EXISTENTIALIZATION** of a sentence with respect to a given name occurring in the sentence is obtained by the following two steps:

1. Replace at least one occurrence of the name in the sentence by a variable α , where α does not already occur in the sentence.
2. Prefix $\exists\alpha$ to the open formula resulting from step 1.

Instance

An **INSTANCE** of a universally or existentially quantified sentence is the result of the following two steps:

1. Remove the initial quantifier.
2. In the open formula resulting from step 1, uniformly replace all occurrences of the unbound variable by a name.

This is called **INSTANTIATING** the sentence.

The name is called the **INSTANTIAL NAME**.

And now back to
our regular
programming

```
def set_mirror_object_to_mirror(modifier_ob, mirror_ob):  
    mirror_mod.mirror_object = mirror_ob
```

```
    operation == "MIRROR_X":  
        mirror_mod.use_x = True  
        mirror_mod.use_y = False  
        mirror_mod.use_z = False  
    operation == "MIRROR_Y":  
        mirror_mod.use_x = False  
        mirror_mod.use_y = True  
        mirror_mod.use_z = False  
    operation == "MIRROR_Z":  
        mirror_mod.use_x = False  
        mirror_mod.use_y = False  
        mirror_mod.use_z = True
```

```
    # Select object at the end add here  
    modifier_ob.select = 1  
    bpy.context.scene.objects.active = modifier_ob  
    modifier_ob.select = 0  
    bpy.context.selected_objects[0].name].select  
    print("please select exactly one object")
```

```
--- OPERATOR CLASSES ---
```

```
    types.Operator):  
        "X mirror to the selected  
        object.mirror_mirror_x"  
        "Mirror X"
```

```
    def execute(self, context):  
        if context.object is not None:
```

New Rules

- Our predicate proof system preserves all the rules from Sentential Logic, and adds four more to deal with the two new types of sentences we introduced earlier in chapter three: universal sentences and existential sentences.
- Thus we will add introduction rules for the universal and existential, and elimination rules for the universal and existential.

Universal Elimination

- How does it work.
- A universal sentence makes a claim about all things. It says that whatever falls within the scope of the universal quantifier is true of all things.
- Thus stating the universal “Everyone who attends TAMU is an Aggie” - $\forall x(Tx \rightarrow Ax)$ would entail the following:
 - If Betty attends TAMU, Betty is an Aggie
 - If David attends TAMU, David is an Aggie
- Insert any name you wish and, if the universal is true, it would apply to any name.

Universal Elimination (one-line rule)

- Given a universally quantified sentence (at line m), conclude any instance of it.
- Annotation: m $\forall E$

1	(1) $\forall x(Gx \rightarrow \sim Sx)$	A
1	(2) $(Ga \rightarrow \sim Sa)$	1 $\forall E$
1	(3) $(Gb \rightarrow \sim Sb)$	1 $\forall E$
1	(4) $(Gc \rightarrow \sim Sc)$	1 $\forall E$

Existential Introduction

- How does it work.
- An existential makes the claim there is *at least one* entity of whom the sentence is true.
- If it is true that “Bob is an Aggie,” would it be true that “Someone is an Aggie?”
- Of course! Bob is a person, so Bob is a “someone.” And if Bob is a someone, then the truth of “Bob is an Aggie (Ab)” would entail the truth of the sentence “Someone is an Aggie ($\exists xAx$).”

Existential Introduction (one-line rule)

- Given a sentence (at line m) containing at least one occurrence of a name, conclude an existentialization of that sentence with respect to that name.
- Annotation: $m \exists I$

1 (1) $Fa \ \& \ Ha$ A

1 (2) $\exists x (Fx \ \& \ Hx)$ 1 $\exists I$

Universal Introduction

- How does it work?
- This one is a little tricky. Remember that a universal sentence is supposed to be true of *all things*. So how can we go from an instance (Fa) to a universal ($\forall xFx$)?
- Well, if we are speaking about a specific someone, but we *could*, in fact, be speaking of *anyone* and the inferences still be true, then it doesn't matter who we are speaking about, simply that we are speaking about a particular entity that satisfies the sentence's truth.
- This rule, thus, requires us to employ a condition to confirm that *any* name could be used and still allow us to arrive at a true claim.

Universal Introduction (one-line rule)

- Given a sentence (at line m) containing at least one occurrence of a name, conclude a universalization of the sentence with respect to that name.
- **Condition:** The name in question must not occur in any assumptions in m 's assumption set.
- Annotation: $m \forall I$

1 (1) $\forall x (Fx \ \& \ Hx)$ A

1 (2) $Fa \ \& \ Ha$ 1 $\forall E$

1 (3) Ha 2 $\&E$

1 (4) $\forall x Hx$ 3 $\forall I$ (check the condition by looking at line 3's assumption set!)

Existential Elimination

- How does it work.
- This one is the trickiest one, mostly because an existential provides us so little information. It tells us that an entity satisfies the listed predicates, but not *which* entity satisfies the set of predicates.
- So to go from “someone is generous” ($\exists xGx$) to “Bob is generous” (Gb) we need to be careful that using “Bob” in our instance doesn’t impact our inferences. We essentially need to treat “Bob” as a variable to represent whichever entity of whom the existential is true.
- So we again need to apply some conditions, to ensure we are not going from something that is true, to something that is false!

Existential Elimination (two-line rule w/ discharge)

- Given a sentence (at line m) and an assumption (at line i) that is an instance of some existentially quantified sentence that is present (at line k), conclude the given sentence again.
- Condition: The instantial name at line i must not appear in the sentence at line k or in the sentence at line m . Also, it must not appear in any of the assumptions belonging to the assumption set at line m , other than the instance i itself.
- Annotation: $k, m \exists E (i)$

Existential Elimination (cont.)

1	(1) $\exists x(Gx \ \& \ \sim Fx)$	A
2	(2) $Ga \ \& \ \sim Fa$	A (for $\exists E$ on 1)
2	(3) Ga	2 &E
2	(4) $\exists xGx$	3 $\exists I$
1	(5) $\exists xGx$	1,4 $\exists E$ (2) (Again, check your conditions!)

- Does the instancial name (a) chosen when we assumed at line 2 occur in the line that motivated the assumption (line 1)?
- Does the instancial name (a) occur in the line whose sentence we repeat (line 4)?
- Does the instancial name (a) occur anywhere in the sentences in the assumption set of the line whose sentence we repeated (line 4)?

$\forall x(Gx \rightarrow Fx), \forall x(Fx \rightarrow Hx) \vdash \forall x(Gx \rightarrow Hx)$

1	(1) $\forall x(Gx \rightarrow Fx)$	A
2	(2) $\forall x(Fx \rightarrow Hx)$	A
3	(3) Ga	A (for $\rightarrow$ I)
1	(4) $Ga \rightarrow Fa$	1 $\forall$ E
1,3	(5) Fa	3,4 $\rightarrow$ E
2	(6) $Fa \rightarrow Ha$	2 $\forall$ E
1,2,3	(7) Ha	5,6 $\rightarrow$ E
1,2	(8) $Ga \rightarrow Ha$	7 $\rightarrow$ I (3)
1,2	(9) $\forall x(Gx \rightarrow Hx)$	8 $\forall$ I

$\exists x(Gx \ \& \ \sim Fx), \ \forall x(Gx \rightarrow Hx) \vdash \exists x(Hx \ \& \ \sim Fx)$

1	(1) $\exists x(Gx \ \& \ \sim Fx)$	A
2	(2) $\forall x(Gx \rightarrow Hx)$	A
3	(3) $Ga \ \& \ \sim Fa$	A (for $\exists E$ on 1)
2	(4) $Ga \rightarrow Ha$	2 $\forall E$
3	(5) Ga	3 $\&E$
2,3	(6) Ha	4,5 $\rightarrow E$
3	(7) $\sim Fa$	3 $\&E$
2,3	(8) $Ha \ \& \ \sim Fa$	6,7 $\&I$
2,3	(9) $\exists x(Hx \ \& \ \sim Fx)$	8 $\exists I$
1,2	(10) $\exists x(Hx \ \& \ \sim Fx)$	1,9 $\exists E$ (3)

$\forall x(Fx \rightarrow \sim Gx) \vdash \sim \exists x(Fx \& Gx)$

1	(1) $\forall x(Fx \rightarrow \sim Gx)$	A
2	(2) $\exists x(Fx \& Gx)$	A (for RAA)
3	(3) $Fa \& Ga$	A (for $\exists E$ on 2)
1	(4) $Fa \rightarrow \sim Ga$	1 $\forall E$
3	(5) Fa	3 $\&E$
1,3	(6) $\sim Ga$	4,5 $\rightarrow E$
3	(7) Ga	3 $\&E$
1,3	(8) $\sim \exists x(Fx \& Gx)$	6,7 RAA (2)
1,2	(9) $\sim \exists x(Fx \& Gx)$	2,8 $\exists E$ (3)
1	(10) $\sim \exists x(Fx \& Gx)$	2,9 RAA (2)